Partition function as a coherent state integral

$$Z = \sum_{\{\alpha\}} e^{-\frac{E_\alpha}{k_B T}} = \sum_{\{\alpha\}} \langle \alpha | e^{-\frac{\hat{H}}{k_B T}} | \alpha \rangle = \mathrm{Tr}\left(e^{-\beta \hat{H}}\right)$$

eigenstates of the system

trace

$$\beta \equiv \frac{1}{k_B T}$$

Generally, we do not know $|\alpha\rangle$ for complicated many-body problems.

Think of β as "time." Start with $\hat{H} = \hat{H}(\hat{S})$

single spin

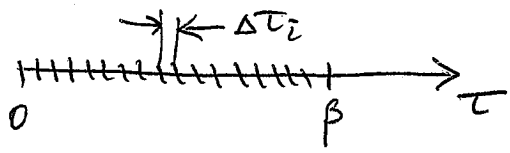
$$Z = \mathrm{Tr}\, \exp(-\Delta\tau_1 \hat{H}) \exp(-\Delta\tau_2 \hat{H}) \ldots \exp(-\Delta\tau_N \hat{H})$$

where $\sum_{i=1}^{N} \Delta\tau_i = \beta$

In each term $(\Delta\tau_i \hat{H})$ the Hamiltonian can be function of "time" as well.

Limit $\Delta\tau_i = \Delta\tau = \frac{\beta}{M}$ for $M \to \infty$

errors introduced by such splitting are small as $(\Delta\tau)^2$.



Now insert operator unity $\hat{I} = \int \underbrace{\mathcal{D}\vec{n}_i}_{\frac{2S+1}{4\pi} d^3\vec{n}_i} |n_i\rangle\langle n_i| = \hat{1}$

at every time τ_i

Note: index (i) here is temporal,

$\vec{n}_i \equiv \vec{n}(\tau_i)$.

Then $Z = \mathrm{Tr} \int \mathcal{D}\vec{n}_1 \, \mathcal{D}\vec{n}_2 \ldots \mathcal{D}\vec{n}_M$ ~~tr~~ .

$$\cdot \, |n_M\rangle\langle n_M| \, e^{-\Delta\tau \hat{H}} \, |n_{M-1}\rangle\langle n_{M-1}| \, e^{-\Delta\tau \hat{H}} \, |n_{M-2}\rangle \ldots$$

$$\ldots \langle n_2| \, e^{-\Delta\tau \hat{H}} \, |n_1\rangle\langle n_1|$$

Trace enforces periodic boundary conditions: $\vec{n}_1 = \vec{n}_M$.

$$\Rightarrow \quad Z = \lim_{M\to\infty} \left(\prod_{j=1}^{M} \int \mathcal{D}\vec{n}_j \right) \langle n_j | e^{-\Delta\tau \hat{H}} | n_{j-1}\rangle =$$

$$= \lim_{M\to\infty} \prod_{j=1}^{M} \int \mathcal{D}\vec{n}_j \, \langle n_j | \, 1 - \Delta\tau \hat{H} \, | n_{j-1}\rangle + O(\Delta\tau)^2.$$

To first order in $\Delta\tau$: $\Delta\tau \langle n_j | \hat{H} | n_{j-1}\rangle =$

$$= \Delta\tau \langle \vec{n}(\tau_j) | \hat{H} | \vec{n}(\tau_j - \Delta\tau)\rangle \approx \Delta\tau \langle \vec{n}(\tau_j) | \hat{H} | \vec{n}(\tau_j)\rangle + O(\Delta\tau)^2$$

While $\langle n_j | n_{j-1}\rangle = \langle n(\tau_j) | n(\tau_j - \Delta\tau)\rangle$

describes overlap of two, consequtive in time, states.

<u>Assume</u> smooth evolution (big assumption, in fact, but it works in most cases)

Recall that $\langle n(\tau+\delta) | n(\tau)\rangle \approx \exp\left[-iS\left(1-\cos\theta(\tau)\right) \frac{d\varphi}{d\tau} \delta \right]$

$$\Rightarrow \langle n_j | n_{j-1}\rangle = \exp\left[-iS\left(1-\cos\theta(\tau_j)\right) \frac{\Delta\varphi(\tau_j)}{\Delta\tau} \Delta\tau \right]$$

The same, in different notation:

$$\langle n(\tau) | n(\tau-\Delta\tau)\rangle \cong \langle n(\tau) | \, n(\tau) - \Delta\tau \frac{d}{d\tau} n(\tau)\rangle =$$

$$= \langle n(\tau)| \left(|n(\tau)\rangle - \Delta\tau \frac{d}{d\tau} |n(\tau)\rangle \right) =$$

$$= \underbrace{\langle n(\tau)|n(\tau)\rangle}_{1} - \Delta\tau \langle n(\tau)| \frac{d}{d\tau} |n(\tau)\rangle$$

$$\Rightarrow \quad Z = \lim_{M\to\infty} \prod_j \int \mathcal{D}\vec{n}(t) \Big[1 - \Delta\tau \Big\{ \langle n(\tau_j)| \frac{d}{d\tau} |n(\tau_j)\rangle +$$

$$+ \langle n(\tau_j)| \hat{H}(\hat{S}) |n(\tau_j)\rangle \Big\}$$

$$\approx \lim_{M\to\infty} \prod_j \int \mathcal{D}\vec{n}(\tau_j) \quad \exp - \Delta\tau \Big\{ \langle n(\tau_j)| \frac{d}{d\tau} |n(\tau_j)\rangle$$

$$+ H(S\vec{n}(\tau_j)) \Big\}$$

$$= \int \mathcal{D}\vec{n}(\tau) \quad \exp - \int_0^\beta d\tau \Big\{ \langle n(\tau)| \frac{d}{d\tau} |n(\tau)\rangle + H(S\vec{n}(\tau)) \Big\}.$$

$$\int_0^\beta d\tau \, \langle n(\tau)| \frac{d}{d\tau} |n(\tau)\rangle = A_B \quad \underline{\text{Berry phase}}$$

From $\underbrace{\frac{d}{d\tau} \langle n(\tau)|n(\tau)\rangle}$ $= \frac{d}{d\tau} (1) = 0$

$$\underbrace{\left\langle \frac{d}{d\tau} n(\tau) \middle| n(\tau) \right\rangle}_{\langle n(\tau)| \frac{d}{d\tau} n(\tau)\rangle^*} + \langle n(\tau)| \frac{d}{d\tau} n(\tau)\rangle = 0$$

$\Rightarrow \quad \mathrm{Re}\left\{ \langle n(\tau)| \frac{d}{d\tau} n(\tau)\rangle \right\} = 0 \Rightarrow A_B$ is purely imaginary.

$$|n\rangle = e^{i\frac{\theta}{2}(\sigma_x \sin\varphi - \sigma_y \cos\varphi)} |\uparrow\rangle = \cos\frac{\theta}{2} + i(\sigma_x \sin\varphi - \sigma_y \cos\varphi)\sin\frac{\theta}{2}$$

$$= \cos\frac{\theta}{2} |\uparrow\rangle + e^{i\varphi} \sin\frac{\theta}{2} |\downarrow\rangle$$

$$A_B = \int d\tau \langle n(\tau)| \frac{d}{d\tau} |n(\tau)\rangle = \int d\tau \langle\uparrow| e^{-i\frac{\theta}{2}\sigma_m} \frac{d}{d\tau} e^{i\frac{\theta}{2}\sigma_m} |\uparrow\rangle$$

$$= \int d\tau \left(\langle\uparrow| \cos\frac{\theta}{2} + \langle\downarrow| e^{-i\varphi} \sin\frac{\theta}{2}\right)\left(-\frac{\dot\theta}{2} \sin\frac{\theta}{2} |\uparrow\rangle + \right.$$

$$\left. + e^{i\varphi} \frac{\dot\theta}{2} \cos\frac{\theta}{2} |\downarrow\rangle + i\dot\varphi e^{i\varphi} \sin\frac{\theta}{2} |\downarrow\rangle\right) =$$

$$= \int d\tau \left(-\frac{\dot\theta}{2} \cos\frac{\theta}{2} \sin\frac{\theta}{2} + \frac{\dot\theta}{2} \sin\frac{\theta}{2} \cos\frac{\theta}{2} + i\dot\varphi \sin^2\frac{\theta}{2}\right)$$

$$= \int d\tau\, i\frac{1}{2}(1-\cos\theta)\,\dot\varphi$$

which is exactly what we had previously.

On the other hand

$$\langle\uparrow| e^{-i\frac{\theta}{2}\sigma_m} \frac{d}{d\tau} e^{i\frac{\theta}{2}\sigma_m} |\uparrow\rangle$$

Seems that

$$\frac{d}{d\tau} e^{i\frac{\theta}{2}(\sigma_x \sin\varphi - \sigma_y \cos\varphi)} =$$

$$= \frac{i}{2}\dot{\theta}\,(\sigma_x \sin\varphi - \sigma_y \cos\varphi)\, e^{i\frac{\theta}{2}(\sigma_x \sin\varphi - \sigma_y \cos\varphi)} +$$

$$+ \; i\frac{\theta}{2}\left(\sigma_x \cos\varphi + \sigma_y \sin\varphi\right)\dot{\varphi}\; e^{i\frac{\theta}{2}(\sigma_x \sin\varphi - \sigma_y \cos\varphi)}$$

But while $\sigma_m\left(\cos\frac{\theta}{2} + i\,\sigma_m \sin\frac{\theta}{2}\right) = \left(\cos\frac{\theta}{2} + i\sigma_m \sin\frac{\theta}{2}\right)\sigma_m$

$$\sigma_{m_2}\, e^{i\frac{\theta}{2}\sigma_{m_1}} = \sigma_{m_2}\left(\cos\frac{\theta}{2} + i\sigma_{m_1}\sin\frac{\theta}{2}\right) \neq$$

$$\left(\cos\frac{\theta}{2} + i\sigma_{m_1}\sin\frac{\theta}{2}\right)\sigma_{m_2}$$

because $\sigma_{m_2}\sigma_{m_1} \neq \sigma_{m_1}\sigma_{m_2}$

(more precisely, $\sigma_{m_2}\sigma_{m_1} = -\sigma_{m_1}\sigma_{m_2}$.)

$\Rightarrow$ taking derivative of an operator requires more care:

$$\frac{d}{d\tau} e^{\hat{O}} = \int_0^1 du\; e^{\hat{O}(1-u)}\, \frac{d\hat{O}}{d\tau}\, e^{\hat{O}u} \qquad (S\ 13.21)$$

For us, $\hat{O} = \frac{i}{2}\theta\left(\hat{\sigma}_x \sin\varphi - \hat{\sigma}_y \cos\varphi\right) = i\frac{\theta}{2}\hat{\sigma}_{m_1}$

Then $A_B = \int d\tau\, \langle n(\tau)|\frac{d}{d\tau}|n(\tau)\rangle = \int d\tau \int_0^1 du \langle\uparrow| e^{-i\frac{\theta}{2}\hat{\sigma}_{m_1}}\, e^{(1-u)i\frac{\theta}{2}\hat{\sigma}_{m_1}} \cdot$

$$\cdot \frac{d}{d\tau}\left[\frac{i}{2}\theta\,\hat{\sigma}_{m_1}\right] e^{u\, i\frac{\theta}{2}\hat{\sigma}_{m_1}} |\uparrow\rangle$$

$$A_B = \frac{i}{2}\int d\tau \int_0^1 du \, \langle \uparrow | \, e^{-\frac{i}{2}u\theta\hat{\sigma}_{m_1}} \left[\dot{\theta}\,\hat{\sigma}_{m_1} + \theta\,\hat{\sigma}_{m_2}\,\dot{\varphi} \right] e^{\frac{i}{2}u\theta\hat{\sigma}_{m_1}} | \uparrow \rangle$$

because $\frac{d}{d\tau}\left(\theta\,\hat{\sigma}_{m_1}\right) = \dot{\theta}\,\hat{\sigma}_{m_1} + \theta\,\frac{d}{d\tau}(\vec{\sigma}\cdot\vec{m}_1) = \dot{\theta}\,\hat{\sigma}_{m_1} + \theta\,\hat{\sigma}_{m_2}\,\dot{\varphi}$

and $\frac{d}{d\tau}\vec{m}_1 = \frac{d}{d\tau}\left(\sin\varphi, -\cos\varphi, 0\right) = \left(\cos\varphi, \sin\varphi, 0\right)\dot{\varphi} = \vec{m}_2\cdot\dot{\varphi}$

We now define $|n(\tau,u)\rangle = e^{\frac{i}{2}u\theta\hat{\sigma}_{m_1}}|\uparrow\rangle$

Note that $\begin{cases} |n(\tau,u=0)\rangle = |\uparrow\rangle \\ |n(\tau,u=1)\rangle = |n(\tau)\rangle \end{cases}$

Also note that $|n(\tau,u)\rangle$ is obtained from $|n(\tau)\rangle$ by extending $\theta \to u\theta$. Hence $\langle n(\tau,u)|\vec{\sigma}|n(\tau,u)\rangle = \vec{n}(\tau,u)$

where the extended unit vector

$$\vec{n}(\tau,u) = \left(\cos\varphi\,\sin(u\theta),\ \sin\varphi\,\sin(u\theta),\ \cos(u\theta)\right).$$

$$\partial_u\vec{n}(\tau,u) = \left(\theta\cos\varphi\cos(u\theta),\ \theta\sin\varphi\cos(u\theta),\ -\theta\sin(u\theta)\right)$$

so that

$$\vec{n}(\tau,u)\times\partial_u\vec{n}(\tau,u) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \cos\varphi\sin(u\theta) & \sin\varphi\sin(u\theta) & \cos(u\theta) \\ \theta\cos\varphi\cos(u\theta) & \theta\sin\varphi\cos(u\theta) & -\theta\sin(u\theta) \end{vmatrix}$$

$$= \hat{x}\left(-\theta\sin\varphi\,\sin^2(u\theta) - \theta\sin\varphi\cos^2(u\theta)\right) +$$

$$-\hat{y}\left(-\theta\cos\varphi\,\sin^2(u\theta) - \theta\cos\varphi\cos^2(u\theta)\right) + \hat{z}\,\underbrace{\cancel{\theta\sin\varphi\cos\varphi\left(\sin u\theta\cos u\theta\right.}}_{=0}$$

$$= -\theta\left(\hat{x}\sin\varphi - \hat{y}\cos\varphi\right) = -\theta\,\vec{m}_1$$

$$\Rightarrow \partial_\tau\left(\vec{n}(\tau,u)\times\partial_u\vec{n}(\tau,u)\right) = -\dot{\theta}\,\vec{m}_1 - \theta\left(\hat{x}\cos\varphi + \hat{y}\sin\varphi\right)\dot{\varphi} =$$

$$= -\dot{\theta}\,\vec{m}_1 - \theta\,\vec{m}_2\,\dot{\varphi} \quad (!)$$

So we can write square-brackets term as

$$[\dot{\theta}\,\hat{\sigma}_{m_1} + \theta\,\hat{\sigma}_{m_2}\,\dot{\varphi}] = -\vec{\sigma}\cdot\partial_\tau\left(\vec{n}(\tau,u)\times\partial_u\vec{n}(\tau,u)\right).$$

Then

$$A_B = -\frac{i}{2}\int d\tau\int_0^1 du\,\langle n(\tau,u)|\,\vec{\sigma}\cdot\partial_\tau\left[\vec{n}(\tau,u)\times\partial_u\vec{n}(\tau,u)\right]|n(\tau,u)\rangle$$

$$= -\frac{i}{2}\int d\tau\int_0^1 du\,\underbrace{\langle n(\tau,u)|\,\vec{\sigma}\,|n(\tau,u)\rangle}_{\vec{n}(\tau,u)}\cdot\partial_\tau\left[\vec{n}(\tau,u)\times\partial_u\vec{n}(\tau,u)\right]$$

$$= -\frac{i}{2}\int d\tau\int_0^1 du\;\vec{n}(\tau,u)\cdot\partial_\tau\left[\vec{n}(\tau,u)\times\partial_u\vec{n}(\tau,u)\right] =$$

$$= -\frac{i}{2}\int d\tau\int_0^1 du\left\{\partial_\tau\left[\vec{n}(\tau,u)\cdot\vec{n}(\tau,u)\times\partial_u\vec{n}(\tau,u)\right]_{=0}\right.$$

$$\left. - \partial_\tau\vec{n}(\tau,u)\cdot\vec{n}(\tau,u)\times\partial_u\vec{n}(\tau,u)\right\} =$$

$$= \frac{i}{2}\int_0^\beta d\tau\int_0^1 du\;\partial_\tau\vec{n}(\tau,u)\cdot\vec{n}(\tau,u)\times\partial_u\vec{n}(\tau,u)$$

$$= i\left(\frac{1}{2}\right)\int_0^\beta d\tau\int_0^1 du\;\vec{n}(\tau,u)\cdot\partial_u\vec{n}(\tau,u)\times\partial_\tau\vec{n}(\tau,u)$$

This is Berry's phase for one spin $\frac{1}{2}$.

For $S > \frac{1}{2}$ $\quad A_B = i\,S\int_0^\beta d\tau\int_0^1 du\;\vec{n}(\tau,u)\cdot\partial_u\vec{n}(\tau,u)\times\partial_\tau\vec{n}(\tau,u)$

It's meaning: area covered by the spin during evolution from $\tau=0$ to $\tau=\beta=\frac{1}{k_BT}$. Note that the boundary condition $\vec{n}(0)=\vec{n}(\beta)$ guarantees that curve swept ~~selected~~ by $\vec{n}(\tau,u)$ is closed

$\Rightarrow$ the choice of $\vec{n}(\tau,u=0) = (0,0,1)$ is a "gauge choice".

Proof of operator identity (S13.21)

$$\frac{d}{d\tau} e^{\hat{O}} = \int_0^1 du\, e^{\hat{O}(1-u)} \frac{d\hat{O}}{d\tau} e^{\hat{O}u}$$

Expand both sides in (infinite) series:

$$LHS = \frac{d}{d\tau} \sum_{n=0}^{\infty} \frac{\hat{O}^n}{n!} = \sum_{n=1}^{\infty} \frac{1}{n!} \frac{d}{d\tau}(\hat{O}^n) = \sum_{n=1}^{\infty} \frac{1}{n!} \left(\dot{O} O^{n-1} + \right.$$

$$\left. + O \dot{O} O^{n-2} + O^2 \dot{O} O^{n-3} + \dots + O^{n-2} \dot{O} O + O^{n-1} \dot{O} \right)$$

where I used the product rule of differentiation:

$$\frac{d}{d\tau}(ABCD\dots) = A' BCD\dots + A B' CD\dots + AB C' D + \dots$$

$$RHS = \int_0^1 du \sum_{m=0}^{\infty} \frac{\hat{O}^m}{m!} (1-u)^m \frac{d\hat{O}}{d\tau} \sum_{l=0}^{\infty} \frac{\hat{O}^l}{l!} u^l$$

Consider term with total of $m+l = n-1$ operators $\hat{O}$:

$$\int_0^1 du \sum_{m,} \frac{\hat{O}^m}{m!} \frac{d\hat{O}}{d\tau} \frac{\hat{O}^{n-1-m}}{(n-1-m)!} (1-u)^m u^{n-m-1}.$$

Do u-integral: $\int_0^1 du\, (1-u)^{m+1-1} u^{n-m-1} = B(m+1, n-m) =$

$$= \frac{\Gamma(m+1)\Gamma(n-m)}{\Gamma(m+1+n-m)} = \frac{m!(n-m-1)!}{n!}$$

$$\Rightarrow \sum_{m=0}^{n-1} \frac{1}{n!} \hat{O}^m \frac{d\hat{O}}{d\tau} \hat{O}^{n-m-1} = \frac{1}{n!} \left(\dot{O} O^{n-1} + O \dot{O} O^{n-2} + \dots \right.$$

$$\left. + O^{n-1} \dot{O} \right)$$

That is, LHS = RHS term by term.